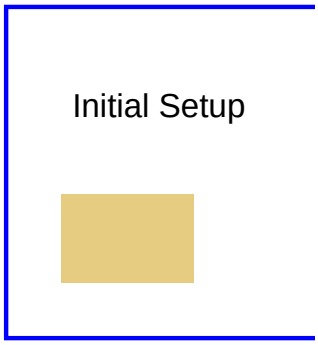

Fig. 2: the overall mechanism.

an alternative approach is to reallocate the task to a suitable core, thereby reducing the impact of parallel resource contention. Under this new allocation, the local and remote blocking of the task would differ entirely from the original allocation.

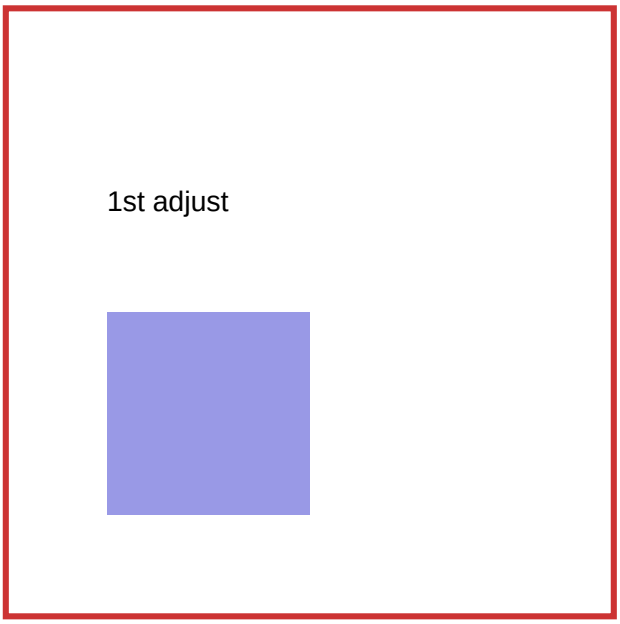

Fig. 3: two priority elevation implementations.

If the task still remains unschedulable after reallocation, the system is deemed unschedulable, leading to the termination of the algorithm without a feasible scheduling solution being found. When multiple tasks miss their deadline, our framework sorts them by priority and optimizes the highest-priority task in each step.